

Econ 30065: Microeconomics

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Choices under uncertainty

Introduction

- Up until now, we assumed that individuals made decisions under ***perfect information***
 - Choices with **certainty**
- In reality, many decision are made under uncertainty
 - A choice has many possible consequences
 - Example: a guy receives as an inheritance the possibility of picking 2 paintings at his uncle house. There is a 1% probability the two paintings are stolen on his way back home, and a 18% probability that 1 painting is stolen.
- How do we make choices when certain variables are **uncertain** (making choices with risk)?

Describing risk

- Risk exists whenever the consequences of a decision are uncertain.
- In economics, we call any situation involving uncertainty a **lottery**
 - **A lottery** is a set of outcomes, each with an associate payoff, and each happening with some probability
- Consumers choose among lotteries → we need to describe them.
 - Example: Alberto is trying to decide whether to buy tickets to a baseball game. His enjoyment of the game depends on an uncertain event beyond his control: whether it rains. He will be happiest if he does not rain.
 - ➔ Buying tickets is a lottery! The possible outcomes are: it rains, it does not rain. The payoff is higher if it does not rain. Probabilities are involved: there is a chance it rains.
- To describe lotteries we must know:
 1. All the possible outcomes
 2. The **probability** or likelihood that each outcome will occur
 3. The **payoffs** of each outcome

Describing risk

- An **outcome** is one possible way in which events relevant to a risky decision can unfold
 - To analyze a risky decision, begin by describing every possible outcome
- ***Outcomes set*** is the set of all possible outcomes
- Once the outcomes of the lottery is realized, the agent experiences only one consequence
 - Each outcome rules out the others

Describing risk

- Some outcomes are more likely than others
- **Probability** is a measure of the likelihood that a given outcome will occur
 - A number between 0 and 1, or a percentage
 - Probability of 0 means an outcome is impossible, probability of 1 means it's certain
 - Add the probabilities of two outcomes to obtain the probability that one of those two outcomes will occur
 - The probabilities of all outcomes always add to 1; it's certain that something will happen
- The **probability distribution** of a set of outcomes gives the likelihood that each possible outcome will occur

Describing risk

- Risky choices often have financial consequences, **payoffs**
- Payoffs can be positive (gains) or negative (losses)
- Every outcome has an associate payoff. Since every outcome happens with some probability, that probability is also the probability of the associate payoff.

Expected value

- To determine the average gain or loss from a lottery, we can calculate its expected value
 - ***Expected value*** (or expected payoff) of a risky choice is a weighted average of all the possible payoffs, using the probability of each payoff as its weight

$$EV = P_1V_1 + P_2V_2 + P_3V_3 + \dots + P_nV_n$$

where P_i is probability of outcome i occurring,

V_i is value of the outcome if state i occurs

Decision making under uncertainty

- Agents choose among lotteries
- We know the expected value of a lottery
- How do agents choose?
 - Depends on the individual
 - Some may be willing to take risk with higher expected value
 - Some will prefer less risk even with lower expected value
- ➔ Decision making under uncertainty
- ➔ Expected utility functions
- ➔ The agent will pick the one that gives him the highest expected utility

Expected Utility Functions

➤ An *expected utility function*:

- Assigns a utility level to each payoff resulting from a possible outcome
- Then take the expected value of those utilities
- It is a weighted average of all possible utility levels using the probability of each level as its weight

Expected Utility maximization

- In general expected utility of a lottery can be expressed as:

$$EU = P_1u(V_1) + P_2u(V_2) + P_3u(V_3) + \dots + P_nu(V_n)$$

where P_i is probability of state i occurring,

V_i is value of the outcome if state i occurs

- This is called Von Neumann- Morgenstern utility function
- The agent will pick the choice that gives him the highest expected utility
- Example: if $U(x) = \sqrt{x}$

$$EU(2\text{paintings}) = 0.81(\sqrt{900}) + 0.18(\sqrt{450}) + 0.01(\sqrt{0}) = 28$$

Expected Utility maximization

- If the guy can choose btw “2 paintings” and “2 necklaces”
 - The outcome “2 necklaces are stolen” happens with probability 0.0025 and has an associated payoff of 0 euro
 - The outcome “1 necklaces are stolen” happens with probability 0.095 and has an associated payoff of 450 euro
 - The outcome “0 necklaces are stolen” happens with probability 0.9025 and has an associated payoff of 900 euro

$$EU(2 \text{ necklaces})=29 > EU(2 \text{ paintings})=28$$

Preferences toward risk

- We see empirically that people differ in their preference toward risk
- People can be *risk averse, risk neutral, or risk loving*

Certainty equivalent

- The ***certainty equivalent*** of a risky bundle is the amount of money which, if provided with certainty, would make the consumer equally well off:

$$U(CE) = E(U(\text{lottery}))$$

Risk premium

- The ***risk premium*** of a risky bundle is the difference between its expected value and the consumer's certainty equivalent:

$$\text{risk premium} = (\text{EV} - \text{CE})$$

- Amount by which a consumer is willing to reduce expected consumption to eliminate all risk